Talagrand's ultimate matching conjecture for finite exponents

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Abstract

We prove the finite-exponent part of Talagrand's ultimate matching conjecture for two independent uniform point clouds in the unit square. For each finite critical pair $1/r_1+1/r_2 = 1/2$, a single matching has bounded coordinatewise exponential r_i-moments at scale $\sqrt{\log N/N}$ with probability tending to one. We reveal the point counts through adaptive anisotropic partitions. A local inverse of divergence controls the fresh current on each child, and an exponential phase with a memory term controls the lag of the routing metric. A terminal correction and an integer relay construction produce a permutation of the original points.

1 Statement and setup

Let $X_1, \dots, X_N$ and $Y_1, \dots, Y_N$ be independent uniform samples in the unit square. Write $s_N = \sqrt{\log N/N}$. The finite-exponent assertion of the ultimate matching conjecture [1] is as follows.

Theorem 1. *Fix $2 \le r_1, r_2 < \infty$ with $1/r_1 + 1/r_2 \ge 1/2$. There is a constant $C = C(r_1, r_2) < \infty$ such that, with probability tending to one as $N \to \infty$, a single permutation π satisfies*

$$\frac{1}{N}\sum_{j=1}^{N} \exp\left[\left(\frac{|X_j^i - Y_{\pi(j)}^i|}{C s_N}\right)^{r_i}\right] \le 2, \qquad i = 1, 2.$$

We prove the critical case $1/r_1 + 1/r_2 = 1/2$, where both exponents exceed two. The strictly subcritical case follows by choosing a coordinatewise larger finite critical pair and reducing the exponents. Indeed $t^r \le 1 + t^{r'}$ for $r \le r'$, and a fixed dilation absorbs the resulting constant in the modular.

Notation. For $p \ge 1$ the Luxemburg norm $\|f\|_{\psi_p}$ is the infimum of $A > 0$ for which the indicated probability space satisfies

$$\int \big(\exp((|f|/A)^p) - 1\big) \le 1.$$

We use Lebesgue measure on the unit square, mass $1/N$ on matched pairs, and mass h^2 per grid face of a given direction, with zeros adjoined to make a probability space in the last case. The triangle

inequality for this norm follows from convexity of the Young function. A current on a domain Q with divergence f and zero normal flux satisfies $\int_Q J \cdot \nabla\varphi = -\int_Q f\varphi$ for smooth tests on the closure, or equivalently after extending both sides by zero off Q. We write $A \preceq B$ for quadratic-form order and $a \lesssim b$ when $a \le Cb$ with C independent of N.

Choose a dyadic width $h = 2^{-m} \asymp s_N$ with sufficiently large fixed lower comparison constant. Then $\mu = Nh^2 \asymp m \asymp \log N$. On the fixed h-grid, all red and blue counts belong to $[c\mu, C\mu]$ with probability $1 - o(1)$, by the binomial bound and a union bound. Denote averaging on this grid by P_h. Set $s = hm^3$, which will be the smallest intrinsic side in the adaptive subdivisions.

We use the following elementary conditional Bernstein estimate. If ξ_j are conditionally independent and centered with $|\xi_j| \le b$ and $\sum_j \mathbb{E}[\xi_j^2 \mid \mathcal{F}] \le v$, then

$$\log \mathbb{E}[\exp(u \textstyle\sum_j \xi_j) \mid \mathcal{F}] \le Cu^2 v \quad (|u|b \le 1), \qquad \mathbb{P}(|\textstyle\sum_j \xi_j| > t \mid \mathcal{F}) \le 2\exp[-c\min(t^2/v, t/b)].$$

For the first bound apply $e^z \le 1 + z + Cz^2$ for $|z| \le 1$ to each summand and multiply the expectations. Optimizing the exponential Markov bound over the permitted tilts gives the second.

2 Finite subdivisions and a local divergence inverse

The following geometric construction will be used in coordinate charts obtained by determinant-one affine maps. Suppose a parallelogram Q has diameter D and contains a ball of radius $c_0 D$, with $c_0 > 0$ fixed. Fix a small $\theta > 0$. If $D \le D_0 s$ we leave Q terminal, where D_0 is a large constant. Otherwise take the maximal squares from an s-dyadic lattice which are contained in Q, have side $2s \le \ell \le \theta D$, and have distance at least $\gamma\ell$ from ∂Q. The distance constant γ is large and fixed. Complete the partition by the intersections of uncovered s-squares with Q. A clipped s-square is assigned to a full uncovered s-square within distance Cs, and these unions are terminal cells. For this assignment, move a point of the clipped square toward the center of the interior ball by a distance $O(s)$ to find a ball of radius $3s$ inside Q. Indeed convexity gives a ball of radius $tc_0 D$ at the point at fraction t along this segment. There is a full s-square in the smaller ball, within distance Cs of ∂Q. Choose γ larger than that constant. The full square is uncovered because any admissible square containing it has boundary distance at least $2\gamma s$. We take D_0 large enough for the central ball construction and for $\theta D_0 > 2$. Each full square receives only $O(1)$ clipped pieces. Empty pieces are discarded, and boundaries of area zero can be assigned by a fixed convention.

The nonterminal children are squares. Assign to each child its side length ℓ, with $\ell = s$ for a merged terminal cell. In the coordinates of Q, each child S satisfies

$$|S| \asymp \ell_S^2, \qquad \operatorname{diam} S \lesssim \ell_S, \qquad \ell_S \asymp \max\{s, \operatorname{dist}(x, \partial Q)\} \quad (x \in S). \tag{1}$$

The constants may depend on c_0, θ, γ, D_0. For a maximal square, its next dyadic ancestor either meets the boundary, fails the distance condition, or exceeds the upper side cap. In the first two cases the ancestor's diameter bounds the additional distance from any point in the child to the boundary. In the third case the child's side is comparable to D. This proves the upper bound on distance in (1), and the distance condition proves the lower bound. The same argument for the $2s$ ancestor of an uncovered full s-square gives boundary distance $O(s)$, since the upper side cap cannot fail. The clipping and assignment preserve this bound. Taking D_0 larger if necessary makes all child diameters at most $C\theta D$. In particular all terminal children from a split have intrinsic diameter $O(s)$. A cell which becomes terminal without a split has intrinsic diameter $O(s)$ in the chart in which termination is decided.

For a shape-regular convex parent Q, take a smooth probability density η_Q supported on a central ball of radius cD. Choose it by translating and dilating a fixed bump. For mean-zero f on

Q define its current by sending $f(y)\,\mathrm{d}y$ along rays from y to z, averaged against $\eta_Q(z)\,\mathrm{d}z$. In a weak pairing this is

$$\int J_f \cdot \nabla\varphi = \int_Q f(y) \int_Q \eta_Q(z)[\varphi(z) - \varphi(y)]\ \mathrm{d}z\ \mathrm{d}y = -\int_Q f\varphi.$$

Thus extension by zero gives exact divergence f and zero normal flux. Its evaluation kernel is

$$K(x,y) = (x-y)\int_1^\infty \eta_Q\big(y + t(x-y)\big)t\ \mathrm{d}t. \tag{2}$$

This formula follows by changing variables from the ray parameter. Direct differentiation gives

$$|K(x,y)| \lesssim |x-y|^{-1}, \qquad |\nabla_x K(x,y)| + |\nabla_y K(x,y)| \lesssim |x-y|^{-2}.$$

On the support of the integrand $t|x-y| \le D$. The bump and its gradient are bounded by CD^{-2} and CD^{-3}, and either differentiation of the bump costs at most a factor t. There is also the stronger support restriction

$$K(x,y) = 0 \qquad \text{if} \qquad |x-y| > C\,\mathrm{dist}(x,\partial Q). \tag{3}$$

Indeed, when $z = y + t(x-y)$ is in the central ball, $x = (1-1/t)y + z/t$. A supporting line realizing the boundary distance of x has distance at least cD from z and nonnegative distance from y. Therefore $\mathrm{dist}(x,\partial Q) \ge cD/t$, whereas $|x-y| \le D/t$.

Lemma 2 (Projected evaluations)**.** *Let P be averaging on the children of Q. For x, x' in the same child S, of size ℓ, and a fixed $0 < \alpha < 1$,*

$$\begin{aligned} \|PK(x,\cdot)\|_2 &\le C, \qquad \|PK(x,\cdot)\|_\infty \le C/\ell, \\ \|P(K(x,\cdot) - K(x',\cdot))\|_2 &\le C(|x-x'|/\ell)^\alpha, \\ \|P(K(x,\cdot) - K(x',\cdot))\|_\infty &\le C\ell^{-1}(|x-x'|/\ell)^\alpha. \end{aligned}$$

The same bounds hold after subtracting the parent averages.

Proof. For either evaluation, (3) and (1) put the support in $B(x, C\ell)$. Every child meeting this support has diameter $O(\ell)$, by the upper distance comparison and the Lipschitz property of boundary distance. Any child meeting $B(x, c\ell)$ has area at least $c'\ell^2$. For a nonterminal S this follows from the lower boundary distance at x, and for a terminal S from the minimum area. Thus the projected support has area $O(\ell^2)$. On its cells near the singularity, the integral bound $O(\ell)$ and the area lower bound give an average of size $O(\ell^{-1})$. Farther away the kernel bound gives the same result.

For $d = |x-x'| \le c\ell$, the integral of the kernel difference on $B(x, C\ell)$ is $O(d[1 + \log(\ell/d)])$. Integrate the two singularities separately within distance $2d$, then use the derivative bound along the segment, which is in Q. Averaging on the near cells divides this bound by $\gtrsim \ell^2$. On the other cells the derivative bound gives $O(d/\ell^2)$ directly. The projected support area and $u(1 + \log(1/u)) \lesssim_\alpha u^\alpha$ prove the claims. Larger d follows from the single-evaluation bounds. Subtracting the parent average contracts L^2 and at most doubles L^∞. □

The same near and far integration, over the whole parent, gives

$$\int_Q |K(x,y) - K(x',y)|\ \mathrm{d}y \lesssim d(1 + \log(D/d)), \qquad d = |x-x'| \le D/2. \tag{4}$$

Thus currents of bounded sources have continuous representatives inside their parents. They extend continuously by zero on the boundary, since (3) gives $\int_Q |K(x,y)|\ \mathrm{d}y \lesssim \mathrm{dist}(x,\partial Q)$.

3 Phase and determinant-one selector

Put $a_i = 1/r_i$, $F_i(t) = (1+t^2)^{r_i/2}$ and

$$H(y) = F_1(y_1) + F_2(y_2), \qquad v = \nabla H, \qquad B(y) = \operatorname{diag}(H^{1-2a_1}, H^{1-2a_2}) + vv^{\mathsf{T}}.$$

Use $C(y) = \sqrt{\det B(y)}\, B(y)^{-1}$. Criticality gives

$$\det B = H(1+W), \qquad W = \sum_i v_i^2 H^{-1+2a_i}, \qquad 1 + W \asymp H.$$

For the last assertion, the upper bound follows from $|v_i| \lesssim H^{1-a_i}$. At large H, at least one $F_i(y_i)$ is at least $H/2$, and its squared derivative is $\gtrsim H^{2-2a_i}$. Bounded states are covered by the term 1. Consequently

$$\det C = 1, \qquad C_{ii} \lesssim H^{2a_i}, \qquad C^{-1} = \chi\left[\operatorname{diag}(H^{-2a_i}) + ww^{\mathsf{T}}\right], \quad w = \frac{v}{\sqrt{H}}, \quad \chi = \sqrt{\frac{H}{1+W}} \asymp 1. \tag{5}$$

For the diagonal estimate, $C_{ii} = B_{jj}/\sqrt{\det B}$ with $j \neq i$, and $B_{jj} \lesssim H^{2-2a_j}$. Then use $1 - 2a_j = 2a_i$.

Let $L(y, G) = \sqrt{\operatorname{Tr}(GC(y)^{-1})}$ for a determinant-one positive definite matrix G. Under the local bounds $G_{ii} \leq MH(y)^{2a_i}$, the lag obeys

$$|\partial_j L(y, G)| \lesssim_M H(y)^{1/2-a_j}. \tag{6}$$

In fact L is the norm of the vector with entries $\sqrt{\chi}\sqrt{G_{ii}}H^{-a_i}$ and $\sqrt{\chi}G^{1/2}w$. One has

$$|\partial_j H| \lesssim H^{1-a_j}, \qquad |\partial_j W| \lesssim H^{1-a_j}, \qquad |\partial_j \chi| \lesssim H^{-a_j}, \qquad |\partial_j w_i| \lesssim H^{1/2-a_i-a_j}.$$

The bounds for W and w follow by differentiating and using $F_i''(y_i) \lesssim H^{1-2a_i}$. The local bounds on the diagonals give $|G^{1/2}u| \leq \sum_i \sqrt{G_{ii}}|u_i|$, proving (6). The same estimate, with a changed constant, holds along segments on which H is comparable to its value at the initial point and where G satisfies the local bounds at that initial point.

For a target C and old matrix G, let G^+ be the following bounded metric update. Fix $K > 1$ and set

$$G^+ = G^{1/2} \operatorname{clip}_{[K^{-1},K]}(G^{-1/2}CG^{-1/2})G^{1/2},$$

where clipping acts on the eigenvalues. The two relative eigenvalues are reciprocal. Hence $\det G^+ = 1$ and $K^{-1}G \preceq G^+ \preceq KG$. The update also satisfies

$$G^+ \preceq K^{-1}G + C. \tag{7}$$

Indeed the scalar clipping function is bounded by $K^{-1} + t$ for $t > 0$. If the target is $C(z)$, contraction of the relative eigenvalue distance gives

$$L(z, G^+) \leq K^{-1/2}L(z, G) + C_K. \tag{8}$$

To see this, if the old relative eigenvalues are e^d, e^{-d}, the new relative eigenvalues with respect to the target are $e^{(d-\log K)_+}, e^{-(d-\log K)_+}$. Taking the square root of their sum gives (8).

4 Revealing empirical counts

At stage k, let $\mathcal{F}_k$ be the history of all labeled cell memberships revealed in both clouds. The next subdivision is chosen using $\mathcal{F}_k$. Use fixed measurable choices for grid placements and ties. Conditional on $\mathcal{F}_k$, the points remain independent and uniform in their respective cells. For the induction step, after a subdivision has been chosen, conditioning an independent parent-uniform position on its child label makes it uniform in that child. Terminal cells keep their counts without further revelation.

Let f_k be the signed count density divided by $\sqrt{N}$ on the partition at stage k. Thus on Q it is $(N_Q^X - N_Q^Y)/(\sqrt{N}|Q|)$. Write $\delta = \epsilon^2/m$, with $\epsilon > 0$ fixed later. We build U_k with divergence $\sqrt{\delta} f_k$ and zero normal flux. At the root $U_0 = 0$ and $G_0 = I$. Write $G_k(x)$ for the metric of the cell containing x at stage k, and suppress the spatial argument when the cell is understood. Each nonterminal parent Q is a square in its own $G_k^{-1/2}$ coordinates. At its center x_Q select the target $C(U_k(x_Q))$, and form G^+ by the bounded update. In the $G^{+\,-1/2}$ chart the parent is a parallelogram of uniform shape regularity, by the relative matrix bound. Use the subdivision of Section 2 in this chart and assign G^+ to all its children. If the parent is too small to split, make it terminal and keep the old field and metric there. For each split, route the fresh source $f_{k+1} - f_k$ inside the parent using (2) in the new chart. It has mean zero in the parent. The determinant-one change back to physical coordinates multiplies the current by $G^{+\,1/2}$ and preserves the divergence under composition with the coordinate map, by a change of variables in the weak identity. Multiply the increment by $\sqrt{\delta}$. The resulting field U_{k+1} has the required divergence.

Choose the child side cap θD so small, in terms of K, that every nonterminal child side is at most θ_0 times the parent side in their respective charts, where $\theta_0\sqrt{K} < 1$. Every nonterminal cell has intrinsic side at least s. Hence each branch terminates in at most $J_0 \le C_{\mathrm{dep}} m$ stages. Pad to this deterministic horizon, doing nothing on terminated branches. All cells have area at least cs^2. In its chart of formation each terminal child has diameter $O(s)$. A parent declared terminal without a split has diameter $O(s)$ in the proposed chart.

All local estimates are used under $N_Q^X, N_Q^Y \le 2N|Q|$. A stronger uniform event is available on the attempted stages. The conditional mean of a new child count is $N_Q|S|/|Q|$. The error threshold $m^{-2}N|S|$ fails with conditional probability at most $2\exp(-cN|S|m^{-4})$ under the parent cap, by Bernstein. Since $N|S| \gtrsim Ns^2 \asymp m^7$, the union failure over at most Cs^{-2} children per stage and $O(m)$ attempted stages is $o(1)$. Starting at the root, on the good count event the density of each cloud differs from N by a relative $O(m^{-1})$. This follows by summing the relative-to-N error m^{-2} along the branch. Count tests may be applied and the construction stopped after each attempted reveal.

In a split parent, let $V_{k+1} = U_{k+1} - U_k$ be the Piola current, and let $Z_{k+1} = (G^+)^{-1/2} V_{k+1}/\sqrt{\delta}$. We use intrinsic coordinates for the estimates on Z_{k+1}. For each test point, Z_{k+1} is a centered sum of independent point contributions. Its evaluation against a direction is the sum of $N^{-1/2}$ times the child-averaged kernel evaluated at the point's label, centered by the parent average, with a sign for the cloud. By Lemma 2 and the parent cap, each directional evaluation has total conditional variance $O(1)$ and summand bound $C/(\sqrt{N}\ell)$ in a child of size ℓ. The corresponding evaluation difference at distance d has variance $O((d/\ell)^{2\alpha})$ and summand bound $C(d/\ell)^\alpha/(\sqrt{N}\ell)$. In particular

$$\log \mathbb{E}[\exp(u \cdot Z_{k+1}(x)) \mid \mathcal{F}_k] \le C|u|^2, \qquad |u| \le c\sqrt{N}s. \tag{9}$$

Lemma 3 (An envelope on each new child)**.** *Fix $0 < \beta < \alpha < 1$. In each child S, there is an envelope $E_{k+1,S}$ for the supremum and the scaled β-Hölder seminorm of Z_{k+1} in intrinsic coordinates.*

For each fixed R, uniformly in old data satisfying the count cap, it has conditional tail

$$\mathbb{P}(E_{k+1,S} > t \mid \mathcal{F}_k) \le C\mathrm{e}^{-ct^2}, \qquad 0 \le t \le R\sqrt{m},$$

for sufficiently large m. The constants c, C can be fixed before choosing R.

Proof. Take nets of mesh $2^{-j}\ell$ inside S with cardinality $O(2^{2j})$. Test pairs at distance $O(2^{-j}\ell)$ in each net, each point against a nearest predecessor, and an anchored value. At most $O(2^{4j})$ tests are needed at level j. At level j use the threshold $D2^{-\alpha j}(1+t+j)$ for differences. Bernstein gives a union failure bounded by $C\mathrm{e}^{-ct^2}$ over all levels, because the exponent before counting pairs is bounded below by

$$c\min\bigl(D^2(1+t+j)^2,\ D\sqrt{N}s(1+t+j)\bigr) \ \ge\ c'D(1+t^2+j)$$

when $t \le R\sqrt{m}$ and m is large. Here D is chosen sufficiently large. Summation of the net increments gives the supremum bound. For two points at distance comparable to $2^{-j}\ell$, sum each chain down to that scale and then use the same-level comparison. The bound on the sum is $C2^{-\beta j}(1+t)$, using $\beta < \alpha$, which proves the seminorm bound. The evaluation field is continuous by (4) applied to the bounded realized density. The construction works on terminal unions since only the diameter bound and the evaluation estimates on the same child are used. Rescaling the threshold by a fixed factor gives the stated tail. □

For a fixed spatial x, write $E_i(x)$ for the envelope of its child at stage i, and use zero on an inactive step. The child is chosen before the fresh labels, so Lemma 3 applies conditionally to $E_i(x)$. Taking R sufficiently large and using the bound on the number of children gives

$$E_{i,S} \le R\sqrt{m} \quad \text{for all attempted stages and all their children} \tag{10}$$

with probability $1 - o(1)$, up to the count cap.

5 Local diagonal bootstrap

We prove by induction on the good history, pointwise on every cell, that

$$G_{k,ii} \le MH(U_k(x))^{2a_i}, \qquad (1-\eta)H(U_{k-1}(x)) \le H(U_k(x)) \le (1+\eta)H(U_{k-1}(x)). \tag{11}$$

The ratio in (11) is asserted after a step, including on terminal children of that step. It is automatic on inactive steps. Here M is a large fixed constant and $\eta > 0$ is small and fixed. The root has $U_0 = 0$ and $G_0 = I$. The selection and partition are well-defined without the local caps. The geometric estimates use the relative bound K only.

For an active parent Q at stage k, the ancestor Hölder bound gives, for $x, z \in Q$,

$$|U_k^i(x) - U_k^i(z)| \le C\sqrt{\delta}\sum_{j=1}^{k}\rho^{k-j}\sqrt{G_{j,ii}(x)}E_j(x), \qquad \rho < 1.$$

Indeed a descendant diameter in the chart of its ancestor child at stage j is at most $C(\theta_0\sqrt{K})^{k-j}$ times the side of that child. This uses side shrink and $G_k \preceq K^{k-j}G_j$. Raising this factor to the Hölder exponent gives ρ^{k-j}. For a newly formed terminal child use its diameter bound, with a fixed constant at the last step. For a frozen leaf use the stage of its last refinement. By induction

$H(U_j(x)) \le (1-\eta)^{-(k-j)} H(U_k(x))$. Choose η small enough that $\rho' = \rho(1-\eta)^{-\max_i a_i} < 1$. Writing $H = H(U_k(x))$, we obtain

$$|U_k^i(x) - U_k^i(z)| \le C\sqrt{\delta M} H^{a_i} \sum_{j=1}^{k} (\rho')^{k-j} E_j(x) \le C\epsilon\sqrt{M} R\, H^{a_i}. \tag{12}$$

For the last inequality use the good envelope history. If $|z_i - y_i| \le bH(y)^{a_i}$ for both coordinates and $b \le 1$, then H is bounded by $CH(y)$ on the segment from y to z, and

$$|H(z) - H(y)| \le CbH(y).$$

The first assertion follows by the power bounds on F_i and the second by integrating the derivatives on the segment. In particular (12) makes the representative phase $H(U_k(x_Q))$ within relative error $O(\epsilon\sqrt{M}R)$ of H.

By (7) and (5),

$$G_{ii}^{+} \le \left[K^{-1}M + C(1 + C\epsilon\sqrt{M}R)^{2a_i}\right] H^{2a_i}.$$

Fix $q_d \in (K^{-1}, 1)$ and M large enough that this is at most $q_d M H^{2a_i}$ for sufficiently small ϵ. Choose η also so that $q_d \le (1-\eta)^{2\max_i a_i}$. On the fresh local envelope cutoff, the increment satisfies

$$|V_{k+1}^i(x)| \le \sqrt{\delta}\sqrt{G_{ii}^{+}}\, E_{k+1}(x) \le C\epsilon\sqrt{M} R\, H^{a_i}.$$

The relative-change estimate above gives the new ratio in (11), by taking ϵ small. The margin q_d then gives $G_{ii}^{+} \le M H(U_{k+1}(x))^{2a_i}$. This proves the induction. Only the fresh local envelope cutoff is required for the pointwise conclusion after an old good history.

6 Drift of the anisotropic phase

For a fixed x in an active parent, put

$$T_k = \sum_{j=1}^{k} \rho_1^{k-j} E_j(x)^2, \qquad L_k = L(U_k(x), G_k), \qquad S_k = L_k^2 + T_k, \qquad \rho' < \rho_1 < 1.$$

The sum in (12) is $O(\sqrt{T_k})$ by Cauchy–Schwarz. Use (8) at the representative, and apply (6) to move between it, $U_k(x)$ and $U_{k+1}(x)$. The local caps hold on these segments up to constants by the relative smallness estimates in the preceding section. Thus on the fresh local cutoff,

$$L_{k+1} \le K^{-1/2} L_k + C + C_M \epsilon \sqrt{H/m}\, (\sqrt{T_k} + E_{k+1}(x)). \tag{13}$$

For example the coordinate error is bounded by $C\sqrt{\delta M} H^{a_i}(\sqrt{T_k} + E_{k+1})$, while the lag derivative in that coordinate costs $C_M H^{1/2-a_i}$.

Use the phase

$$\Phi_k = \lambda_k H(U_k)(1 + \kappa\delta S_k), \qquad \lambda_k = \mathrm{e}^{-\Lambda k\delta},$$

with spatial modular overshoot threshold T_{stop} fixed. Freeze S_k on inactive steps. Before stopping, $H \le C_H m$ uniformly. Indeed (12) on the current leaf makes H at every point of that leaf comparable to H at any given point of it. A leaf has area $\gtrsim s^2$ and $\lambda_k \ge 1/2$ will be arranged. The modular bound gives $H \le C_H m$, with C_H fixed. For old frozen leaves use the estimate when they were formed. This argument uses the inductive local caps and does not require a preexisting global matrix cap.

By (13), squaring with fixed slack and updating $T_{k+1} = \rho_1 T_k + E_{k+1}^2$ gives

$$S_{k+1} \le q_0 S_k + C_0(1 + E_{k+1}^2), \qquad q_0 < 1. \tag{14}$$

One can choose q_0 strictly between $\max(K^{-1}, \rho_1)$ and 1, and then take ϵ small. In the initial choices take ρ_1 fixed above the original geometric rate ρ, then take η small so that $\rho' < \rho_1$ and $(1+\eta)q_0 < 1$. Since $H' \le (1+\eta)H$ on the local cutoff,

$$H'S_{k+1} - HS_k \le -cHS_k + CH(1 + E_{k+1}^2).$$

There is no additional higher power from the memory product.

Write $y = U_k(x)$ and $V = V_{k+1}(x)$. On the local cutoff, H stays comparable to its old value along the segment from y to $y + V$. Since each $F_i'' \lesssim H^{1-2a_i}$ along that segment, Taylor gives

$$H(y+V) - H(y) \le v(y) \cdot V + C \sum_i H^{1-2a_i} |V_i|^2.$$

Write $\lambda = \lambda_{k+1}$, $H = H(U_k)$, $E = E_{k+1}(x)$, $Z = Z_{k+1}(x)$ and $\ell = \lambda\sqrt{\delta}(G^+)^{1/2}v(y)$. Since $vv^{\mathsf{T}}/H \preceq C_1 C(y)^{-1}$ and $G^+ \preceq KG_k$,

$$|\ell|^2 \lesssim \lambda^2 \delta H L_k^2.$$

By (11), $L_k^2 \lesssim_M H$, so the conditional linear MGF at 2ℓ is within its allowed range $c\sqrt{N}s$. The Taylor quadratic term is bounded by $C_M \lambda\delta H E^2$. It follows that on the local cutoff

$$\Phi_{k+1} - \Phi_k \le -(\lambda_k - \lambda)H - c\kappa\lambda\delta H S_k + C\kappa\lambda\delta H + \ell \cdot Z + uE^2, \qquad u \lesssim_{M,\kappa} \lambda\delta H.$$

Since $H \le C_H m$, u is small for sufficiently small ϵ. Cauchy–Schwarz separates the conditional linear MGF from the truncated quadratic factor. The latter uses

$$\mathbb{E}[\mathrm{e}^{2uE^2}\mathbf{1}_{E \le R\sqrt{m}} \mid \mathcal{F}_k] \le 1 + \int_0^{R\sqrt{m}} 4ut\mathrm{e}^{2ut^2}\mathbb{P}(E > t \mid \mathcal{F}_k)\, \mathrm{d}t \le 1 + Cu$$

for $2u < c$ bounded away from c. The combined log cost is $O(|\ell|^2 + u)$, hence $O(\lambda\delta H S_k) + O(\lambda\delta H)$. Choose κ large to pay the first term, then Λ large to pay the second from $\lambda_k - \lambda \ge \Lambda\lambda\delta$. The conditional modular decreases on the cutoff. The constants can be chosen in this order. Fix the geometry, choose $q_d > K^{-1}$, choose $\rho_1 > \rho$, and choose $q_0 > \max(K^{-1}, \rho_1)$, all below one. Choose η for the three strict margins in the oscillation, cap and memory estimates. Fix M large enough for the cap margin assuming the representative phase ratio is at most two. Then choose $\kappa, \Lambda, T_{\mathrm{stop}}$. Use $\lambda_k \ge 1/2$ provisionally for the bootstrap C_H. Choose R for the conditional union bound and finally take ϵ small enough for every smallness constraint, including $\Lambda J_0 \epsilon^2/m \le \log 2$. The relative oscillation estimate ensures the representative ratio bound once ϵ is small.

Define the scalar process by the spatial modular while active. On count or envelope failure set it to zero and keep it there. On modular overshoot keep the attained value thereafter. At a point on a terminal branch, the phase decreases because the field and memory freeze and the clock decreases. Conditional integration of the pointwise bound proves the supermartingale property because the simultaneous fresh good event has indicator at most the local envelope cutoff used in the pointwise estimate. The initial modular is bounded by a fixed constant, since $H(0) = 2$, $G_0 = I$, and the initial lag is fixed. A nonnegative supermartingale starting at B_0 crosses T_{stop} with probability at most B_0/T_{stop}, by stopping at first crossing. Choose T_{stop} large and fixed. Count and envelope failures have probability $o(1)$, so the probability of completion with bounded modular is bounded below. On this event $\int \exp(\frac{1}{2}|U_{J_0}^i|^{r_i}) \le T_{\mathrm{stop}}$. Jensen applied to a small power of this exponential gives a fixed Luxemburg bound for $U_{J_0}^i$. The unnormalized leaf current $J_{\mathrm{leaf}} = U_{J_0}/\sqrt{\delta}$ therefore has the two coordinate norms $O(\sqrt{m})$.

7 Correction to the fixed grid

On success the diagonal caps and $H \lesssim m$ give $\|G_k\| \lesssim m^{2a_*}$, where $a_* = \max_i a_i$. This also bounds the proposed metric in which a parent is declared terminal without a split. The physical terminal leaf diameters are at most $D_* = Csm^{a_*}$. Let f_{leaf} be their signed count density divided by $\sqrt{N}$, and let $f_{\mathrm{emp}} = N^{-1/2}(\sum_j \delta_{X_j} - \sum_j \delta_{Y_j})$. The leaf current has exact divergence f_{leaf}. To reach the fixed histogram $P_h f_{\mathrm{emp}}$ we add currents for

$$(P_h - I)f_{\mathrm{leaf}}, \qquad P_h(f_{\mathrm{emp}} - f_{\mathrm{leaf}}).$$

For the first source use the averaged-ray inverse on each h-square, where the source has mean zero. The L^1 kernel bound yields current supremum at most $Ch\|f_{\mathrm{leaf}}\|_\infty$. The relative count event gives $\|f_{\mathrm{leaf}}\|_\infty \lesssim \sqrt{N}/m$, so this is $o(\sqrt{m})$.

Use the averaged-ray inverse on the whole unit square for the second source. Denote its evaluation kernel by K_x, a function of the source variable. The test against a source atom uses $P_h K_x$, which satisfies

$$|P_h K_x(z)| \lesssim (h + |x - z|)^{-1}.$$

This follows by integrating the $1/|x - z|$ singularity over the cell when $|x - z| = O(h)$, and the kernel bound away from it. Condition on the full terminal revelation, including the adaptive tree and success. Each point is independent and uniform in its leaf. Its contribution to the correction is centered by the leaf average of $P_h K_x$, with magnitude at most $C/(h\sqrt{N})$ after normalization. By the leaf density cap, the total evaluation variance is bounded by

$$C \sum_{Q\ \mathrm{leaf}} \int_Q |P_h K_x - (P_h K_x)_Q|^2.$$

For leaves meeting $B(x, C(D_* + h))$, bound by the squared regularized kernel on a slightly larger ball, at cost $C(1 + \log_+(D_*/h))$. On the other leaves, the source-variable derivative bound for K_x gives oscillation $O((D_* + h)/r^2)$ at distance r from x. Indeed each pair of grid cells meeting one leaf is at most $D_* + O(h)$ apart, so the same bound holds for their averages. The squared integral over this far region is $O(1)$. Thus the variance is bounded by a deterministic $v = C(1 + \log m)$.

For the correction current J_{corr}, Bernstein gives the conditional pointwise tail

$$\mathbb{P}(|J_{\mathrm{corr}}(x)| > t \mid \mathcal{F}_{J_0}) \le C \exp\bigl[-c \min(t^2/v, h\sqrt{N}\, t)\bigr]$$

on successful histories. On the fixed-grid occupancy event its source density is bounded by $C\sqrt{N}$. The evaluation modulus (4) then shows that a net of mesh N^{-2} controls the supremum, up to an $o(1)$ error. At threshold $R_1\sqrt{mv}$ with large fixed R_1, Bernstein and a union bound give $\|J_{\mathrm{corr}}\|_\infty = O_{\mathbb{P}}(\sqrt{mv})$ on old success. Up to that cutoff the conditional pointwise tail is bounded by $C\exp(-ct^2/v)$ for large N. Tail integration gives a bounded expected truncated spatial quadratic exponential modular at scale $C\sqrt{v}$. Markov's inequality and a fixed dilation of a bounded modular give $\|J_{\mathrm{corr}}\|_{\psi_2} = O_{\mathbb{P}}(\sqrt{v})$ on the supremum event. If $\|J\|_\infty \le A$ and $\|J\|_{\psi_2} \le B$, then $\|J\|_{\psi_r} \le A^{1-2/r}B^{2/r}$ for $r \ge 2$. At this scale the r-th power is bounded pointwise by $(|J|/B)^2$. It follows that for each fixed finite r_i the correction has ψ_{r_i} norm $O_{\mathbb{P}}(m^{1/2-1/r_i}\sqrt{\log m}) = o_{\mathbb{P}}(\sqrt{m})$. All corrections have exact divergence and zero normal flux. Together with the leaf construction they give, with probability bounded below, a current for $P_h f_{\mathrm{emp}}$ with both coordinate norms $O(\sqrt{m})$.

8 From the current to a single point permutation

Write $\mu = Nh^2$, $g = R - B$ for the raw count difference on the fixed grid, and orient its faces. The grid divergence is net outgoing load. We record the transfer and probability upgrade so the proof ends on the original labeled points.

Extraction of a face flow

In one dimension let I_0 be linear interpolation through the cell centers, constant on the two outer half-cells. Write D_0 for the unscaled forward difference on center values. The map from center values to cell averages is $B_0 = I - D_0^{\mathsf{T}} D_0/8$. The identity follows by integrating each hat on the two half-cells, with the constant extension at the ends. Put $B_1 = I - D_0 D_0^{\mathsf{T}}/8$, so $D_0 B_0 = B_1 D_0$. Both inverse matrices have absolute row and column sums at most 2 by the Neumann series.

For a grid test u, apply I_0 in both coordinates to $(B_0^{-1} \otimes B_0^{-1})u$. The resulting test has cell averages u. Its horizontal derivative is h^{-1} times the interpolation of $(B_1^{-1} \otimes B_0^{-1})D_1 u$, constant between centers in the horizontal variable and linear in the transverse variable. The vertical derivative has the exchanged formula. Each product basis function for the derivatives has mass h^2, and the basis functions sum to at most one. For a direction i, write β_e for these basis functions and T_i for the inverse tensor filter. For a current J of divergence $P_h f_{\mathrm{emp}} = g/(\sqrt{N}h^2)$, put

$$A_{i,e} = h^{-2} \int J_i \beta_e, \qquad q_i = \sqrt{N} h \, T_i^{\mathsf{T}} A_i.$$

The weak divergence equation yields $\sum_i \langle q_i, D_i u \rangle = -\langle g, u \rangle$. For faces oriented in the forward direction this means $-\sum_i D_i^{\mathsf{T}} q_i = g$, the net outgoing divergence. These piecewise bilinear tests are Lipschitz and may be used in the weak identity by extending and mollifying. In the Jensen estimate, dividing the absolute entries of a filter by its maximal row sum gives subprobability weights. The column sum bound controls the sum over faces. Thus each coordinate face norm is at most $C\sqrt{N}h$ times the continuum coordinate norm. Each face of a fixed direction carries mass h^2 in this Luxemburg norm, with zeros adjoined if needed. Consequently the raw count load scale is $\sqrt{N}h\sqrt{m} \asymp \mu$.

Upgrade to high probability

Let $Q(g)$ be the minimum, over real flows of divergence g, of the maximum of the two coordinate face ψ_{r_i} norms. This is a convex norm on mean-zero grid sources and is uniformly Lipschitz in their Euclidean norm. In fact the elementary square grid isoperimetric inequality $|\partial S| \gtrsim \sqrt{\min(|S|, |S^c|)}$ and Cauchy–Schwarz give $|\sum_{v \in S} g_v| \le C\|g\|_2 |\partial S|$. For the isoperimetric inequality, if neither a row nor a column of S is full, all occupied rows and columns are mixed, and their counts have product at least $|S|$. If some row is full and another is empty, every column is mixed. If there is a full row and no empty row, the complement has neither a full row nor a full column. The case of a full column follows by exchanging coordinates. A flow with uniform capacity $C\|g\|_2$ exists by these cut inequalities. Indeed give each edge two directed arcs with that capacity, attach a source to the positive divergences and a sink to the negative ones, and maximize the total sent. A maximizing flow exists by compactness. If the demand has not been met, the vertices reachable from the source in the residual network define a cut of capacity below the demand. A residual augmenting path would increase the flow, so none exists. This contradicts the cut inequalities. Subtract opposite arc loads to obtain the desired signed flow. The bound on its norm and the triangle inequality prove the Lipschitz bound.

For a replacement of one point by an independent uniform point, let g' be the new source and choose a supporting vector u for Q at g in the mean-zero subspace. Then $\|u\|_2 \le C$ and $(Q(g) - Q(g'))_+^2 \le 2(u_{\text{old}}^2 + u_{\text{new}}^2)$ by convexity. Sum over the $2N$ replacements. The old-cell terms are bounded by the maximum pooled occupancy times C. The fresh-cell terms cost $O(Nh^2)$ after conditional expectation. The expected maximum pooled occupancy is $O(\mu)$ since the exponential moment of the count in one cell is at most $\exp(C\mu)$ at tilt one. Jensen thus bounds the expected maximum by $\log(h^{-2}) + C\mu$. For independent inputs W_j and a function F of them, the reveal martingale variance decomposition and conditional Jensen give $\operatorname{Var} F \le \sum_j \mathbb{E}[\operatorname{Var}(F \mid W_{-j})]$, where W_{-j} denotes all inputs except j. To see this, the j-th martingale difference is the conditional expectation, given $W_1, \dots, W_j$, of $F - \mathbb{E}[F \mid W_{-j}]$. Each conditional variance equals half the mean squared change on resampling W_j. By exchangeability this half equals the mean squared positive change. Applying this to Q with the bounds above gives $\operatorname{Var} Q = O(\mu)$. Fixed positive success at $C\mu$ implies $\mathbb{E}Q \le C\mu + O(\sqrt{\mu})$ by Chebyshev, then $Q \le C'\mu$ with probability tending to one. The optimization can be performed directly on the observed terminal counts.

Relays and individual pairs

On the fixed-grid occupancy event each cell has between $c\mu$ and $C\mu$ points of each color. Round each load between its original floor and ceiling while preserving integer divergence. For this, the subgraph of fractional edges has no vertex of degree one, since the total divergence is integral at every vertex. If this subgraph is nonempty it contains a cycle. Change the loads by a signed constant along that cycle until one hits an endpoint of its allowed interval, and repeat. The face gauges increase by at most a constant. Remove directed cycles and decompose the resulting integer flow into unit paths from surplus to deficit cells. Reserve the corresponding red and blue points at the endpoints. In cell v, let $M_v = \min(R_v, B_v)$ be the remaining number of possible red-blue relays. For internal visits entering in coordinate direction i, let $T_{v,i}$ count them and give each weight $w_{v,i} = \min(1, M_v/(2T_{v,i}))$ when $T_{v,i} > 0$.

In the ordered list of internal visits of each path, group successive visits until their weight first reaches one, then repeat. Each completed group has weight less than two. Choose one relay in each completed group. Fractionally this is feasible by assigning a relay proportionally to the visit weights, because the total weight offered by cell v is at most M_v. To make this choice integral, use a network from groups of demand one to the eligible cells of capacity M_v. The fractional choice shows that all demand can be met. With integer capacities, augmenting paths from zero can use integer increments until demand is met, by the residual cut argument above. A relay uses a blue point for the incoming segment and a red point for the outgoing segment. Pair the points along the resulting segments, and pair all leftover points within their own cells.

Between successive relays or endpoints, the internal visits have total weight at most four. The coordinate displacement in units of h is at most two plus the number of internal visits entering in that coordinate direction. For $\Psi(t) = \exp(t^{r_i}) - 1$ and $A > 0$, Jensen with denominator six therefore charges the modular at displacement scale $6hA$ to a constant $\Psi(1/A)$ and to $w_{v,i}\Psi(1/(Aw_{v,i}))$ over these visits. Each cell is charged total weight $O(\mu)$. Moreover $1/w_{v,i}$ is bounded by a constant times one plus the sum of its incoming direction-i face magnitudes divided by μ. Since $\mu/N = h^2$, the Luxemburg triangle inequality gives the point-coordinate norm bound

$$\|\Delta_i\|_{\psi_{r_i},\text{points}} \le Ch(1 + \|q_i\|_{\psi_{r_i},\text{faces}}/\mu).$$

Here the load sum at a cell involves at most two adjacent direction-i faces, and fixed modular prefactors are absorbed by convexity under a fixed dilation. Both coordinates refer to the same

permutation. Since $h \asymp s_N$ and the optimized face norms are $O(\mu)$ with high probability, this proves Theorem 1.

References

[1] M. Talagrand, *My favorite problems*, manuscript, 2026.

[2] `https://github.com/grottofrancesco/UltimateMatchingTheorem`.